

Delivery Atlas

Target: 0:00 - 0:30 (30 seconds).

Presenter cue: Introduce the project and distinguish the working data lab from the proposed app.

SAY

Our project is Delivery Atlas, an analytics infrastructure for a fictional food delivery app inspired by Uber Eats and Glovo. We want to collect navigation events and show them in a dashboard. We already have a working ClickHouse lab with large public datasets.

The delivery app itself is our next development step.

TRANSITION

First, let me show you how the current system works.

IF ASKED - AFTER THE TALK

Are these Uber Eats data?

No. The app is fictional. OTTO and REES46 provide historical e-commerce events. Ele.me provides food recommendation samples, and Glovo provides a catalogue.

Current architecture and tables

Target: 0:30 - 1:15 (45 seconds).

Presenter cue: Point to the table from input to daily rollup. Explain the role of each layer.

SAY

The system has a web frontend, a Python API and one ClickHouse node running in Docker. Python scripts validate source files and load data in batches. The database stores detailed data in separate tables: OTTO contains events, while Ele.me contains labelled recommendation samples.

We also keep daily counts in a rollup table for faster queries. The API queries ClickHouse and returns JSON to the dashboard. Our replay loads old events now, so we preserve both the original event time and the ingestion time.

TRANSITION

This working lab is the basis for our proposed app event model.

IF ASKED - AFTER THE TALK

Why keep the datasets separate?

They represent different populations and record types. They do not share valid user or product keys, so combining them into one journey would misrepresent the data.

Proposed platform event model

Target: 1:15 - 2:30 (75 seconds).

Presenter cue: Explain the field groups, then point to PARTITION BY and ORDER BY. Say clearly that this model is proposed.

SAY

In the proposed model, each row represents one app event, such as opening a restaurant or adding a product to the cart. We store an event ID, a session ID, timestamps, the action and its context.

The provenance fields distinguish simulated traffic, new demo actions and historical sources. ClickHouse stores data by columns, which suits analytical queries that read selected fields across many rows.

MergeTree is the storage engine. PARTITION BY groups data by month within the node.

ORDER BY sorts the data and supports a sparse primary index. We chose this order for filters by source, date, city and event type, but it still needs benchmarking. Neither the sorting key nor a UUID prevents duplicate submissions.

The schema exists as SQL, but it is not deployed yet.

TRANSITION

Our current lab already shows why a daily rollup can be useful.

IF ASKED - AFTER THE TALK

Is PARTITION BY the same as sharding?

No. These partitions group data within the current node. Sharding distributes data across nodes. ORDER BY controls physical sorting and does not enforce uniqueness.

Historical data vs daily rollup

Target: 2:30 - 3:20 (50 seconds).

Presenter cue: Point to the two bars. Read the rounded times, then explain the method and the scope of the result.

SAY

We compared two ways of answering the same question: how many OTTO events occurred each day, by event type? The historical table contains about 217 million events. Its query took about 5.89 seconds. The daily rollup returned the same answer in about 0.21 seconds, a 27.9 times speedup.

It contains 87 combinations of day and event type. We used the median of three local runs with the SQL query cache disabled. This demonstrates the benefit of pre-aggregation for this query on our machine.

TRANSITION

Performance is one part of the project. We also need to explain its guarantees.

IF ASKED - AFTER THE TALK

Does this prove ClickHouse is faster than PostgreSQL?

No. Both queries ran in ClickHouse. The experiment compares historical data and a rollup for one query. It does not compare engines or measure p99 latency.

Current guarantees and limitations

Target: 3:20 - 4:15 (55 seconds).

Presenter cue: Use the table to explain the limitations as specific next tasks. Emphasize duplicates and recovery.

SAY

The course theory helps us explain the limits of this version. We have one node and persistent storage, but no replicas or high availability. Local data partitions do not demonstrate horizontal scaling. Another issue is event delivery: retrying the current POST request can create duplicate rows.

We need to test retries with a stable event ID and make sure the rollup does not count them twice. We also need to measure ingestion rate, dashboard delay and latency percentiles under load. These are the next steps for validating the infrastructure.

TRANSITION

These limitations define our next milestone.

IF ASKED - AFTER THE TALK

What does idempotent ingestion mean?

Submitting the same logical event again should not change the result beyond the first submission. We must verify both the detailed event count and the rollup count after retries.

Next milestone

Target: 4:15 - 5:00 (45 seconds).

Presenter cue: Finish with the acceptance criteria. Do not open the dashboard during the five-minute talk unless asked.

SAY

Our next milestone is to connect a small delivery app and a reproducible traffic generator to the proposed event model. We will check the complete flow: a user action produces an event, ClickHouse stores it, and the dashboard displays the aggregate.

A repeated submission must not increase the count twice. We will measure the workload and reproduce the setup on another machine. The current lab already provides enough data volume to support this work.

TRANSITION

Thank you. We can show the current dashboard or discuss the database design.

IF ASKED - AFTER THE TALK

Why not add Kafka immediately?

We first need to define and measure the ingestion requirements. A durable stream could help with replay and independent consumers, but it adds operational work and does not by itself eliminate duplicates.

Rehearsal and key facts

Six slides. Five minutes. Timings are rehearsal targets.

Read SAY and TRANSITION during rehearsal. Use the presenter cue to decide what to point to. Save IF ASKED for discussion after the talk.

628,425,832 is the combined OTTO and REES46 row count from separate populations. It is not a unified clickstream.

Ele.me contributes 2,170,299 loaded recommendation samples. Glovo contributes catalogue data.

Replay ingests historical events now. New dashboard button clicks are stored separately.

The benchmark uses 5.8908 and 0.2113 seconds before rounding. The 87 groups combine day and event type.

The proposed app_events schema is not yet deployed. Persistence, replication and sharding are different properties.

The cover is a conceptual AI-generated illustration, not a diagram of the deployed infrastructure.

PROJECT REFERENCES

docs/ESTRUCTURA-BD.md - database structure

docs/RESULTADOS.md - measurements and conditions

docs/REVISION-TEORIA.md - review against course notes

docs/AVANCE-PRESENTACION.md - slide notes and sources